Promising Directions Assessment for `p-vs-np`

Neuro-symbolic AI Group
Idiap Research Institute

March 16, 2026

Abstract

This report identifies the most promising mathematical directions currently surfaced by the framework for `p-vs-np`. The assessment is content-first: each route is described by its theorem pair, typed bridge object, preserved invariant, missing route lemma, and failure logic. The novelty claim is assessed at the level of mathematical program rather than theorem announcement. The conclusion is that the current run contains real route-level novelty and a plausible discovery direction, but not a validated theorem-level discovery.

1 Run Context

- Generated (UTC): 2026-03-16T22:26:32.766903+00:00
- Source run: 2026-03-16T22:26:28.893056+00:00
- Evidence count: 0
- Unresolved hypotheses: 22

2 Executive Summary

There is real programmatic novelty in the sense of a new content-level bridge program: the framework isolates specific theorem-pair deltas, proposes typed intermediate carriers, and exposes missing route lemmas. There is not yet evidence of theorem-level novelty, because the decisive lemmas remain open and the run has not accumulated external or adversarial evidence.

3 Most Promising Directions

3.1 1. MCSP witness family: major consequences such as NP not in P/poly

- Bridge ID: `bridge.001`
- Type: magnification bridge
- Priority: 0.744
- Route: `st.def.hardness-magnification-phenomenon-programme-leve` → `st.thm.oliveira-pich-santhan`
- Abstraction: MCSP witness family
- Carrier: family of instances equipped with a monotone witness measure

- Invariant: MCSP complexity profile in nonuniform-circuits
- Route synopsis: Move from 'st.def.hardness-magnification-phenomenon-programme-leve' on problem 'MCSP' in model 'nonuniform-circuits' using 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' toward 'st.thm.oliveira-pich-santhanam-magnification-near-state' in model 'nonuniform-circuits' to obtain 'major consequences such as NP not in P/poly'.

Why It Is Promising

- Bridge priority 0.744 with sketch score 0.905, mathematical specificity 1.000, evidence signal 0.188, and sketch operationalization 1.000.
- The route synopsis is: Move from 'st.def.hardness-magnification-phenomenon-programme-leve' on problem 'MCSP' in model 'nonuniform-circuits' using 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' toward 'st.thm.oliveira-pich-santhanam-magnification-near-state' in model 'nonuniform-circuits' to obtain 'major consequences such as NP not in P/poly'.
- The proof family 'invariant-lift-through-intermediate-object' is used because Introduce a conjectured intermediate object or invariant, prove that the source regime induces it, then force the target consequence through that invariant.
- Linked hypotheses: 4 with verdict profile {'supported': 3, 'inconclusive': 1} and best confidence 0.804.

Content-wise Novelty Argument

- The route is theorem-pair specific: it tries to move from 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' to 'major consequences such as NP not in P/poly' rather than proposing a generic topic-level bridge.
- The candidate novelty is the object 'MCSP witness family', typed as 'family of instances equipped with a monotone witness measure', which is meant to make the hidden transport mechanism mathematically explicit.
- The route is constrained by the preserved invariant 'MCSP complexity profile in nonuniform-circuits', so the proposal is not only to reach the target consequence but to preserve a specific mathematical profile while doing so.
- The novelty remains programmatic rather than theorem-level because the decisive missing lemma is still: Prove that a hardness-magnification witness family turns 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' into 'major consequences such as NP not in P/poly' while preserving 'MCSP complexity profile in nonuniform-circuits'.
- The route includes an explicit falsification probe: Allow 'MCSP witness family' to ignore the source-side content 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' and only preserve topical overlap. bridge-degrades-or-fails The bridge object must absorb the actual source-side mathematical content rather than merely name the antecedent.

Key Obligations

- Prove the source-to-object claim: the source-side content 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' available at 'st.def.hardness-magnification-phenomenon-programme-leve' can be organized into 'MCSP witness family' as a family of instances equipped with a monotone witness measure.
- Prove the payoff claim: once 'MCSP witness family' carries the route data from 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' toward 'major consequences such as NP not in P/poly', it forces 'NP-not-in-P/poly' at 'st.thm.oliveira-pich-santhanam-magnification-near-state'.
- Prove the transport claim: every admissible realization of 'MCSP witness family' preserves 'MCSP complexity profile in nonuniform-circuits' while crossing the route nonuniform-circuits.
- Discharge the route lemma: Prove that a hardness-magnification witness family turns 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' into 'major consequences such as NP not in P/poly' while preserving 'MCSP complexity profile in nonuniform-circuits'.

Main Risks

- Failure mode: 'MCSP witness family' never becomes more than topical overlap because the route-specific lemma remains missing. Prove that a hardness-magnification witness family turns 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' into 'major consequences such as NP not in P/poly' while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Abort logic: Abort if the negative-route intervention does not expose the expected failure: Allow 'MCSP witness family' to ignore the source-side content 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' and only preserve topical overlap. => bridge-degrades-or-fails.
- Global limitation: evidence_count=0 and unresolved hypotheses=22, so the route is still under mathematical pressure.

Expert Actions

- Discharge the route-specific lemma: Prove that a hardness-magnification witness family turns 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' into 'major consequences such as NP not in P/poly' while preserving 'MCSP complexity profile in nonuniform-circuits'.
- Stabilize the main bridge lemma: Source-to-object lemma: the source-side content 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' at 'st.def.hardness-magnification-phenomenon-programme-leve' yields a realization of 'MCSP witness family' as a family of instances equipped with a monotone witness measure, satisfying admits source-side witness data for 'A barely-nontrivial lower bound for an explicit problem Q in a weak model'; preserves the route invariant 'MCSP complexity profile in nonuniform-circuits'; forces the target-side payoff 'NP-not-in-P/poly'.
- Run the designated counterfactual probe: Allow 'MCSP witness family' to ignore the source-side content 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' and only preserve topical overlap. bridge-degrades-or-fails The bridge object must absorb the actual source-side mathematical content rather than merely name the antecedent.

3.2 2. MCSP witness family: major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter...

- Bridge ID: `bridge.002`
- Type: magnification bridge
- Priority: 0.732
- Route: `st.def.hardness-magnification-phenomenon-programme-leve` → `st.thm.oliveira-santhanam-ha`
- Abstraction: MCSP witness family
- Carrier: family of instances equipped with a monotone witness measure
- Invariant: MCSP complexity profile in nonuniform-circuits
- Route synopsis: Move from 'st.def.hardness-magnification-phenomenon-programme-leve' on problem 'MCSP' in model 'nonuniform-circuits' using 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' toward 'st.thm.oliveira-santhanam-hardness-magnification-for-na' in model 'formula-size' to obtain 'major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter...'.

Why It Is Promising

- Bridge priority 0.732 with sketch score 0.897, mathematical specificity 1.000, evidence signal 0.188, and sketch operationalization 1.000.
- The route synopsis is: Move from 'st.def.hardness-magnification-phenomenon-programme-leve' on problem 'MCSP' in model 'nonuniform-circuits' using 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' toward 'st.thm.oliveira-santhanam-hardness-magnification-for-na' in model 'formula-size' to obtain 'major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter...'.
- The proof family 'invariant-lift-through-intermediate-object' is used because Introduce a conjectured intermediate object or invariant, prove that the source regime induces it, then force the target consequence through that invariant.
- Linked hypotheses: 4 with verdict profile {'supported': 3, 'inconclusive': 1} and best confidence 0.804.

Content-wise Novelty Argument

- The route is theorem-pair specific: it tries to move from 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' to 'major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter...' rather than proposing a generic topic-level bridge.
- The candidate novelty is the object 'MCSP witness family', typed as 'family of instances equipped with a monotone witness measure', which is meant to make the hidden transport mechanism mathematically explicit.

- The route is constrained by the preserved invariant 'MCSP complexity profile in nonuniform-circuits', so the proposal is not only to reach the target consequence but to preserve a specific mathematical profile while doing so.
- The novelty remains programmatic rather than theorem-level because the decisive missing lemma is still: Prove that a hardness-magnification witness family preserves 'MCSP complexity profile in nonuniform-circuits' while transporting the route from the nonuniform-circuits regime to the formula-size regime strongly enough to derive 'major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter...'.
- The route includes an explicit falsification probe: Allow 'MCSP witness family' to ignore the source-side content 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' and only preserve topical overlap. bridge-degrades-or-fails The bridge object must absorb the actual source-side mathematical content rather than merely name the antecedent.

Key Obligations

- Prove the source-to-object claim: the source-side content 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' available at 'st.def.hardness-magnification-phenomenon-programme-leve' can be organized into 'MCSP witness family' as a family of instances equipped with a monotone witness measure.
- Prove the payoff claim: once 'MCSP witness family' carries the route data from 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' toward 'major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter...', it forces 'NP-not-in-P/poly' at 'st.thm.oliveira-santhanam-hardness-magnification-for-na'.
- Prove the transport claim: every admissible realization of 'MCSP witness family' preserves 'MCSP complexity profile in nonuniform-circuits' while crossing the route nonuniform-circuits $\rightarrow$ formula-size.
- Discharge the route lemma: Prove that a hardness-magnification witness family preserves 'MCSP complexity profile in nonuniform-circuits' while transporting the route from the nonuniform-circuits regime to the formula-size regime strongly enough to derive 'major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter...'.

Main Risks

- Failure mode: 'MCSP witness family' never becomes more than topical overlap because the route-specific lemma remains missing. Prove that a hardness-magnification witness family preserves 'MCSP complexity profile in nonuniform-circuits' while transporting the route from the nonuniform-circuits regime to the formula-size regime strongly enough to derive 'major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter...'.
- Abort logic: Abort if the negative-route intervention does not expose the expected failure: Allow 'MCSP witness family' to ignore the source-side content 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' and only preserve topical overlap. => bridge-degrades-or-fails.

- Global limitation: evidence_count=0 and unresolved hypotheses=22, so the route is still under mathematical pressure.

Expert Actions

- Discharge the route-specific lemma: Prove that a hardness-magnification witness family preserves 'MCSP complexity profile in nonuniform-circuits' while transporting the route from the nonuniform-circuits regime to the formula-size regime strongly enough to derive 'major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter...'.
- Stabilize the main bridge lemma: Source-to-object lemma: the source-side content 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' at 'st.def.hardness-magnification-phenomenon-programme-leve' yields a realization of 'MCSP witness family' as a family of instances equipped with a monotone witness measure, satisfying admits source-side witness data for 'A barely-nontrivial lower bound for an explicit problem Q in a weak model'; preserves the route invariant 'MCSP complexity profile in nonuniform-circuits'; forces the target-side payoff 'NP-not-in-P/poly'.
- Run the designated counterfactual probe: Allow 'MCSP witness family' to ignore the source-side content 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' and only preserve topical overlap. bridge-degrades-or-fails The bridge object must absorb the actual source-side mathematical content rather than merely name the antecedent.

3.3 3. MCSP witness family: There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for...

- Bridge ID: `bridge.016`
- Type: magnification bridge
- Priority: 0.711
- Route: `st.def.hardness-magnification-phenomenon-programme-leve` → `st.thm.atserias-m-uller-simp`
- Abstraction: MCSP witness family
- Carrier: family of instances equipped with a monotone witness measure
- Invariant: exponential-lower-bound witness complexity
- Route synopsis: Move from 'st.def.hardness-magnification-phenomenon-programme-leve' on problem 'MCSP' in model 'nonuniform-circuits' using 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' toward 'st.thm.atserias-m-uller-simple-general-magnification-fr' to obtain 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for...'.

Why It Is Promising

- Bridge priority 0.711 with sketch score 0.890, mathematical specificity 1.000, evidence signal 0.200, and sketch operationalization 1.000.

- The route synopsis is: Move from 'st.def.hardness-magnification-phenomenon-programme-leve' on problem 'MCSP' in model 'nonuniform-circuits' using 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' toward 'st.thm.atserias-m-uller-simple-general-magnification-fr' to obtain 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for...'.
- The proof family 'invariant-lift-through-intermediate-object' is used because Introduce a conjectured intermediate object or invariant, prove that the source regime induces it, then force the target consequence through that invariant.
- Linked hypotheses: 5 with verdict profile {'supported': 4, 'inconclusive': 1} and best confidence 0.804.

Content-wise Novelty Argument

- The route is theorem-pair specific: it tries to move from 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' to 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for...' rather than proposing a generic topic-level bridge.
- The candidate novelty is the object 'MCSP witness family', typed as 'family of instances equipped with a monotone witness measure', which is meant to make the hidden transport mechanism mathematically explicit.
- The route is constrained by the preserved invariant 'exponential-lower-bound witness complexity', so the proposal is not only to reach the target consequence but to preserve a specific mathematical profile while doing so.
- The novelty remains programmatic rather than theorem-level because the decisive missing lemma is still: Prove that a hardness-magnification witness family upgrades the route from major separation consequence to exponential lower bound without losing 'exponential-lower-bound witness complexity'.
- The route includes an explicit falsification probe: Allow 'MCSP witness family' to ignore the source-side content 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' and only preserve topical overlap. bridge-degrades-or-fails The bridge object must absorb the actual source-side mathematical content rather than merely name the antecedent.

Key Obligations

- Prove the source-to-object claim: the source-side content 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' available at 'st.def.hardness-magnification-phenomenon-programme-leve' can be organized into 'MCSP witness family' as a family of instances equipped with a monotone witness measure.
- Prove the payoff claim: once 'MCSP witness family' carries the route data from 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' toward 'There exist general-purpose magnification theorems (not problem-specific) showing that certain mild lower bounds for...', it forces 'exponential-lower-bound' at 'st.thm.atserias-m-uller-simple-general-magnification-fr'.
- Prove the transport claim: every admissible realization of 'MCSP witness family' preserves 'exponential-lower-bound witness complexity' while crossing the route nonuniform-circuits.

- Exhibit the model-alignment claim: 'MCSP witness family' gives a faithful translation across nonuniform-circuits.

Main Risks

- Failure mode: 'MCSP witness family' never becomes more than topical overlap because the route-specific lemma remains missing. Prove that a hardness-magnification witness family upgrades the route from major separation consequence to exponential lower bound without losing 'exponential-lower-bound witness complexity'.
- Abort logic: Abort if the negative-route intervention does not expose the expected failure: Allow 'MCSP witness family' to ignore the source-side content 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' and only preserve topical overlap. => bridge-degrades-or-fails.
- Global limitation: evidence_count=0 and unresolved hypotheses=22, so the route is still under mathematical pressure.

Expert Actions

- Discharge the route-specific lemma: Prove that a hardness-magnification witness family upgrades the route from major separation consequence to exponential lower bound without losing 'exponential-lower-bound witness complexity'.
- Stabilize the main bridge lemma: Source-to-object lemma: the source-side content 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' at 'st.def.hardness-magnification-phenomenon-programme-leve' yields a realization of 'MCSP witness family' as a family of instances equipped with a monotone witness measure, satisfying admits source-side witness data for 'A barely-nontrivial lower bound for an explicit problem Q in a weak model'; preserves the route invariant 'exponential-lower-bound witness complexity'; aligns the model shift (nonuniform-circuits) with an explicit transport map.
- Run the designated counterfactual probe: Allow 'MCSP witness family' to ignore the source-side content 'A barely-nontrivial lower bound for an explicit problem Q in a weak model' and only preserve topical overlap. bridge-degrades-or-fails The bridge object must absorb the actual source-side mathematical content rather than merely name the antecedent.

4 Novelty Assessment

4.1 Positive Evidence

- The leading route is theorem-grounded at the level of source and target excerpts, not only labels: st.def.hardness-magnification-phenomenon-programme-leve → st.thm.oliveira-pich-santhanam-magnification-near-state.
- The bridge objects are typed carriers such as 'MCSP witness family' with carrier 'family of instances equipped with a monotone witness measure', rather than generic placeholders.
- The routes expose explicit missing lemmas and negative routes, so novelty is attached to a falsifiable mathematical program rather than to generic thematic similarity.

4.2 Limits

- The current run still has evidence_count=0, so support is structural and explanatory, not evidential.
- There remain 22 inconclusive hypotheses, so the frontier is not yet mathematically settled.
- The novelty is therefore programmatic: new route decompositions, typed carriers, and missing lemmas, not validated new theorems.

4.3 Verdict

There is real programmatic novelty in the sense of a new content-level bridge program: the framework isolates specific theorem-pair deltas, proposes typed intermediate carriers, and exposes missing route lemmas. There is not yet evidence of theorem-level novelty, because the decisive lemmas remain open and the run has not accumulated external or adversarial evidence.

5 Next Best Direction To Explore

- Bridge ID: `bridge.002`
- Title: MCSP witness family: major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter...
- Route: `st.def.hardness-magnification-phenomenon-programme-leve` → `st.thm.oliveira-santhanam-ha`
- Assessment: This is the next best direction because it keeps the leading bridge object fixed while forcing it through a harder transport regime. If the same witness family survives this adjacent target, that is evidence that the carrier is mathematically real rather than tuned only to one theorem statement.
- Missing route lemma: Prove that a hardness-magnification witness family preserves 'MCSP complexity profile in nonuniform-circuits' while transporting the route from the nonuniform-circuits regime to the formula-size regime strongly enough to derive 'major lower bounds for NP in much stronger models (including superpolynomial formula/circuit lower bounds, and in some parameter...'.

6 Recommended Probe

- Hypothesis: `phyp.p-vs-np.021`
- Target statement: `st.thm.oliveira-santhanam-hardness-magnification-for-na`
- Priority: 0.690
- Recommended action: run-another-abduce-deduce-round
- Reason: high priority inherited from connector hypotheses